

ST BENEDICT'S COLLEGE



SUBJECT
GRADE
EXAMINER
NAME
CLASS

AP Mathematics – Stats
11
Mr Benecke
Memo

DATE
MARKS
MODERATOR
DURATION

19 June 2017
80
Mrs Povall and Mrs Sillman
1 hour

QUESTION	MAXIMUM MARK	ACTUAL MARK
1	12	
2	13	
3	10	
4	9	
5	5	
6	9	
7	15	
8	7	
	80	

QUESTION 1**12 MARKS**

Susan found a chart showing her height in cm from 1 year to 3 years' old, recorded every 3 months, as shown below:

Age in months (x)	12	15	18	21	24	27	30	33	36
Height in cm (y)	76	79	82	85	87	89	92	94	96

- a) Determine the value of the correlation coefficient and comment on the correlation of the data. Round off to 3 decimal places:

4

$$r = 0,997 \checkmark$$

Correlation is linear ✓, positive ✓, and very strong (perfect) ✓

- b) Determine the equation of the regression line.

4

$$y = 0,83x \checkmark\checkmark + 66,8 \checkmark\checkmark$$

- c) Determine what Susan's expected height would have been at birth and comment on the reliability of your answer?

4

$$y = 0,83(0) + 66,8 \checkmark$$

$$y = 66,8\text{cm} \checkmark$$

Unreliable ✓, data is extrapolated. ✓ or

Reliable ✓, correlation is almost perfectly linear ✓

QUESTION 2**9 MARKS**

Given $P(A) = 0,8$; $P(B') = 0,25$ and $P(A \cup B)' = 0,05$.

a) Prove that $P(A \cap B) = 0,6$

4

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \quad \checkmark$$

$$P(B) = 0,75 \quad \text{and} \quad P(A \cup B) = 0,95 \quad \checkmark$$

$$0,95 = 0,8 + 0,75 - P(A \cap B) \quad \checkmark$$

$$P(A \cap B) = 0,60 \quad \checkmark$$

b) Determine whether events A and B are independent, show all calculations and give a reason for your answer.

3

$$P(A) \cdot P(B) = 0,8 \times 0,75 = 0,60 \quad \checkmark$$

$$\therefore \text{Independent} \quad \checkmark$$

$$P(A \cap B) = P(A) \cdot P(B) \quad \checkmark$$

c) Determine $P(A|B)$.

2

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$P(A|B) = \frac{0,6}{0,75} \quad \checkmark$$

$$P(A|B) = 0,8 \quad \checkmark$$

QUESTION 3**13 MARKS**

- a) The digits 0, 1, 2, 3, 4, 5, 6, 7, 8, and 9 are used to make 4 digit passwords, digits may be repeated:

- (i) How many different passwords are possible? (Hint: codes may start with 0) 1

$$10^4 = 10000 \checkmark$$

- (ii) What is the probability that a password will be greater than 6000? 3

$$\frac{4 \times 10 \times 10 \times 9 \checkmark}{10000 \checkmark} = \frac{9}{25} \checkmark$$

- (iii) What is the probability that a password will be divisible by 2? 3

$$\frac{10 \times 10 \times 10 \times 5 \checkmark}{10000 \checkmark} = \frac{1}{2} \checkmark$$

- b) The letters from the word BARCELONA are arranged in a line; the repeated letters are treated as identical. Determine:

- (i) The total number of possible arrangements 2

$$\frac{9! \checkmark}{2! \checkmark} = 181440$$

- (ii) The probability that the word will begin with "B" and end with "L" 4

$$\frac{7! \div 2! \checkmark \checkmark}{181440 \checkmark} = \frac{1}{72} \checkmark$$

QUESTION 4**10 MARKS**

5 boys and 4 girls are to be seated randomly in a row, what is the probability that:
(round off to 4 decimal places if necessary)

- a) All the girls are sitting next to each other.

3

$$P(\text{girls next to each other}) = \frac{4! \times 6!}{9!} = \frac{1}{21} \text{ or } 0,0476$$

- b) All the boys sit next to each other, given that all the girls are sitting next to each other.

4

$$P(\text{girls next to each other} \cap \text{boys next to each other}) = \frac{2! \times 4! \times 5!}{9!} = \frac{1}{63} \text{ or } 0,0159$$

$$P(\text{boys next to each other} | \text{girls next to each other}) = \frac{0,0159}{0,0476} = \frac{1}{3} \text{ or } 0,33$$

- c) The boys and girls sit in alternate positions.

3

$$\frac{5! \times 4!}{9!} = \frac{1}{126} \text{ or } 0,0079$$

QUESTION 5**5 MARKS**

2% of all the vehicles that Toyota manufactures are defective.

(round off to 4 decimal places if necessary)

- a) If you bought a Toyota yesterday, what is the probability that it is defective?

1

$$2\% \text{ or } \frac{1}{50} \checkmark$$

- b) A random sample of 50 vehicles are tested. Determine the probability that 2 of the vehicles are defective .

4

$$\checkmark \quad \checkmark \quad \checkmark$$
$$\binom{50}{2} (0,02)^2 (0,98)^{48} = 0,1858 \checkmark$$

QUESTION 6

9 MARKS

Charlie, Mike, Dave, John and Mark are playing soccer, they decide to draw straws (one at a time) to determine which one of them will be the Goalkeeper. The person that draws a short straw first will be the Goalkeeper. Charlie's wife holds out the ten straws for the men to take. Seven straws are the same size, while three of the straws are shorter than the others. Note: The straws are held out in such a way that the men cannot see the length of the straw until they have taken one.

- a) The order that they draw straws is as follows: Mark, Mike, Dave, John and then Charlie. What is the probability that Charlie will be the Goalkeeper.

6

✓ ✓ ✓ ✓ ✓ ✓

$$\frac{7}{10} \times \frac{6}{9} \times \frac{5}{8} \times \frac{4}{7} \times \frac{3}{6} = \frac{1}{12}$$

- b) If Charlie wants to be the goalkeeper, what advice would you give Charlie so that he has the highest probability of being the goalkeeper. Calculate the probability of Charlie being the goalkeeper if he follows your advice:

3

Charlie should be the first person to draw straws ✓✓ The probability will be $\frac{3}{10}$ ✓

QUESTION 7**15 MARKS**

A shipping company owns 150 trucks, of those 150 trucks 25 are red and 30 are blue. If a random sample of 10 trucks are selected, what is the probability that:

(round off to 4 decimal places if necessary)

- (i) 3 trucks are red?

4

$$\frac{\binom{25}{3}\binom{125}{7}}{\binom{150}{10}} = 0,1568$$

- (ii) 3 trucks are red and 4 trucks are blue

5

$$\frac{\binom{25}{3}\binom{30}{4}\binom{95}{3}}{\binom{150}{10}} = 0,0075$$

- (iii) At least 2 trucks are red

6

$$1 - \left(\frac{\binom{25}{0}\binom{125}{10}}{\binom{150}{10}} + \frac{\binom{25}{1}\binom{125}{9}}{\binom{150}{10}} \right) = 0,5215$$

QUESTION 8

7 MARKS

The nine individual letters of the word BARCELONA are placed in a bag. You reach your hand in and select four simultaneously. How many different results will you be able to achieve?

Case 1: No A's ${}^7C_4 = 35$ ✓✓

Case 2: 1 A ${}^7C_3 = 35$ ✓✓

Case 3: 2A's ${}^7C_2 = 21$ ✓✓

$35 + 35 + 21 = 91$ ✓